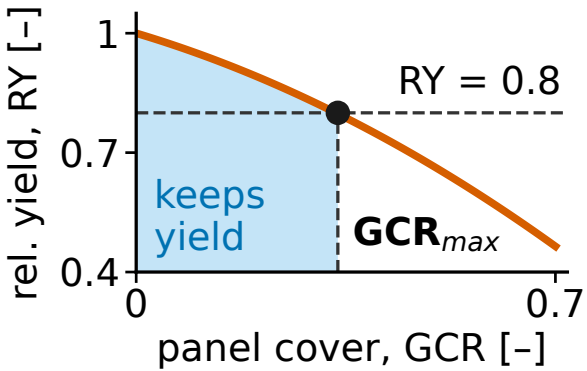
Solar vs food on farmland

competing for land

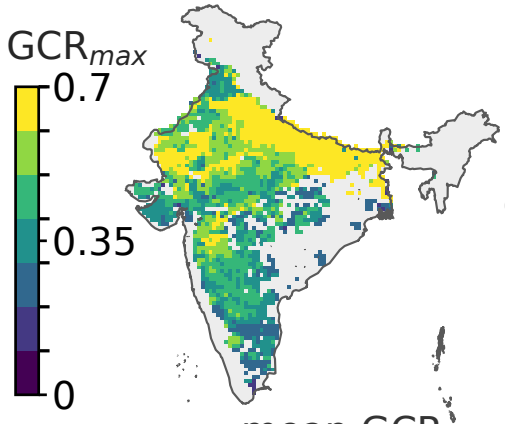
Simulate shade → yield

shade model → crop model



simulated for every 0.25° cell
4 crops · 3 layouts
5 climate models

Where agrivoltaics keeps ≥ 80% yield



45.5 TWp

upper bound

east-west, slope ≤ 5%
GCM range 45.5–45.9

2050: [±X]%

SSP2-4.5, median

mean GCR$_{max}$ per crop

wheat	chickpea	pearl millet	groundnut
0.62	**0.27**	**0.58**	**0.32**